Does Variate Attention Pay Off? Probing Cross-Channel Information in Multivariate Time-Series Foundation Models

Anonymous

Anonymous Institution
anonymous@placeholder.org

Abstract. Recent zero-shot time-series foundation models (TSFMs) increasingly integrate cross-channel attention, claiming superiority over channel-independent baselines. Yet, whether and how these models utilize inter-channel dependencies remains an unsettled empirical question. We present the first systematic diagnostic probing protocol for cross-variate attention in pretrained TSFMs, evaluated on Google's 330M-parameter TimesFM-3. Across 8 standard benchmarks (9,600 test windows), we reveal a pronounced domain bimodality: while variate attention aids physically constrained systems (e.g., weather +8.0%, electricity +2.9%), it causes severe degradation on high-frequency noise domains (solar −32.0%). Through 11 targeted causal interventions, we show that TimesFM-3 is strictly permutation-invariant, vulnerable to uncorrelated noise (20.1% error surge), and relies on micro-phase synchronization rather than static feature pooling. Evaluations against exact Kalman filter oracles on synthetic state-space models quantify the limits of zero-shot factor disentanglement. Finally, mapping the accuracy-latency frontier across channel budgets $K \in \{1..128\}$ reveals that correlation-based Top-K selection capped at 32 channels achieves optimal Pareto efficiency on RTX 4090 GPUs.

Keywords: Time-Series Foundation Models · Multivariate Forecasting · Variate Attention · Diagnostic Probing · Empirical Evaluation.

1 Introduction

The field of time-series forecasting has shifted from task-specific architectures toward zero-shot foundation models (TSFMs) [6,21]. A central architectural divergence among contemporary foundation models concerns channel communication. Early univariate models treated each variate independently [6]. In contrast, next-generation architectures such as TimesFM-3 [9] and Chronos-2 [2] introduce specialized cross-channel attention mechanisms, reporting superior aggregate rankings across diverse benchmark suites.

However, recent empirical evidence presents conflicting observations regarding cross-channel modeling. While TimesFM-3 highlights multivariate gains, Moirai-2.0 [14] explicitly reverted from any-variate cross-attention to independent

univariate forecasting, citing scalability and robustness concerns. Concurrently, Liu et al. [16] (Falcon-X) observed that Chronos-2 demonstrates near-identical error rates with and without group attention on GIFT-Eval, exhibiting attention map collapse across dissimilar inputs. While Falcon-X leveraged this single-benchmark finding to motivate a new architecture, whether Google's 330M-parameter TimesFM-3 suffers from similar idling, how cross-channel attention behaves under controlled interventions, and whether its gains stem from genuine dynamic synchronization remain unknown. In the supervised regime, Abdelmalak et al. [1] and Han et al. [10] demonstrated that channel-independent (CI) architectures frequently outperform channel-dependent (CD) models due to weak inter-channel dependence in standard benchmarks and capacity-robustness trade-offs under distribution shifts.

These contradictory findings motivate a fundamental research inquiry: *Does variate attention truly pay off in zero-shot foundation models, and what mechanisms govern its behavior?* To move beyond aggregate benchmark scores, we propose a within-model diagnostic probing framework evaluated on TimesFM-3. Our study addresses four concrete research questions:

1. **RQ1 (Benefit Profile)**: Does multivariate mode consistently outperform univariate mode across horizons and domains, or do domain-specific boundaries emerge?
2. **RQ2 (Mechanistic Attribution)**: Does the model actively extract synchronized temporal alignments, or does it merely rely on marginal statistics and shared diurnal phases?
3. **RQ3 (Theoretical Gap Closure)**: When ground-truth cross-channel couplings exist, how much theoretically available information does the zero-shot model capture relative to exact Kalman filter oracles?
4. **RQ4 (Channel Budget & Efficiency)**: How does performance scale with the channel budget K, and how should channels be partitioned in high-dimensional database settings?

Contributions. Our primary contributions are summarized as follows:

- **C1: Diagnostic Probing Protocol**: We introduce a unified evaluation protocol quantifying Benefit (B), Reliance (R), Sensitivity (S), and Oracle Gap Closure (GC) through paired interventions on identical windows.
- **C2: Empirical Discovery of Domain Bimodality**: On 8 standard benchmarks across 4 horizons (Table 2), we demonstrate that variate attention is not uniformly beneficial. It delivers steady gains on physical systems (e.g., Weather, Electricity) but suffers extreme degradation on high-frequency, non-stationary domains (Solar error degrades by -31.95%).
- **C3: Causal Mechanism Discovery**: Using 11 target-centric interventions, we prove that TimesFM-3 possesses strict permutation invariance, relies heavily on micro-phase alignment (via IAAFT surrogates), and exhibits acute vulnerability to distractor noise channels.
- **C4: Practical Channel Budget Guidelines**: We map the accuracy-latency Pareto frontier on an RTX 4090, proving that correlation-driven

Top-K channel selection capped at 32 variates preserves peak accuracy while halving inference latency.

2 Related Work

2.1 Multivariate Time-Series Foundation Models

The progression of time-series foundation models has seen diverging architectural paradigms regarding variate interactions. Early architectures, including TimesFM [6] and Chronos [3], operated primarily under channel independence, treating multivariate series as decoupled batches of univariate signals. Recently, TimesFM-3 [9] introduced full cross-variate attention layers interwoven after temporal self-attention in each of its 20 transformer layers, reporting improved aggregate rankings on benchmarks such as GIFT-Eval [8], fev-bench [11], and TIME [20]. Similarly, Chronos-2 [2] introduced group attention across series sharing the same patch index, relying on synthetic multivariatization during pretraining.

However, the efficacy of cross-channel modeling in foundation models remains contested. Moirai [21] adopted any-variate attention by flattening all variables into a single sequence with learned binary variate embeddings; yet its successor, Moirai-2.0 [14], discarded cross-variate attention entirely, forecasting each variate independently to avoid negative transfer and scalability bottlenecks. Other architectures, such as Toto-2.0 [7] and TiRex-2 [4], explore asymmetric attention where targets attend to covariates without reciprocal connections. Crucially, Liu et al. [16] (Falcon-X) observed that Chronos-2 produces near-identical predictions on GIFT-Eval whether group attention is enabled or disabled, displaying attention map collapse across dissimilar inputs. While Falcon-X treated this observation as motivation to introduce an alternative architecture, our work establishes the first systematic within-model diagnostic probing framework for TimesFM-3, dissecting when cross-variate attention provides genuine value, when it idles, and when it severely degrades predictions.

2.2 Channel Independence vs. Channel Dependence Controversy

In the supervised time-series literature, the tension between channel-independent (CI) and channel-dependent (CD) modeling has sparked extensive debate. Landmark studies such as PatchTST [17] and DLinear [24] demonstrated that simple CI architectures routinely outperform sophisticated cross-channel transformers (e.g., Informer [26], Autoformer [23], FEDformer [27]). Although iTransformer [15] revisited channel dependence by embedding entire univariate series as tokens, Han et al. [10] established that CI models enjoy superior robustness under distribution drift, formalizing a capacity-robustness trade-off.

Furthermore, Abdelmalak et al. [1] revealed that the apparent superiority of CI models on standard benchmarks stems from weak inter-channel Granger causality and untuned lookback windows; on standard benchmarks, CI won 21

out of 28 comparisons, whereas on coupled physical ODE systems, CD models won 18 out of 24. Song et al. [19] introduced CLOC, demonstrating that channel communication must be dynamically regulated: channel suppression is optimal on ETT datasets, whereas local-global communication proves essential on Electricity and Solar. Similarly, CPiRi [25] showed that supervised CD models easily overfit channel ordering, while LIFT [22] highlighted that lead-lag relationships represent the primary source of cross-channel gain. Our study investigates whether these exact structural phenomena persist in zero-shot foundation models without task-specific training.

2.3 Probing and Interpretability in Time-Series

While diagnostic probing is well established in natural language processing and computer vision, its application to time-series foundation models remains in its infancy [28,5]. A known limitation of analyzing raw attention maps is that attention weights do not necessarily reflect true causal explanations [12]. To avoid this trap, we implement input-level causal perturbation protocols. Specifically, we leverage Iterative Amplitude Adjusted Fourier Transform (IAAFT) surrogates [18] to randomize cross-channel phase spectra while strictly preserving autocorrelation and marginal amplitude distributions. In parallel, we introduce synthetic linear Gaussian state-space systems with exact analytical Kalman filter oracles, providing a mathematically grounded reference to quantify theoretical mutual information extraction.

3 Diagnostic Probing Framework

3.1 Probing Metrics

Let $\mathcal{W}$ denote the evaluation window set, H the forecast horizon, and $i \in \{1, \ldots, N\}$ the target variate index. For each window $w \in \mathcal{W}$ and target i, the model produces three forecast outputs: the univariate forecast $\hat{y}^{\mathrm{UV}}_{w,i}$, the standard multivariate forecast $\hat{y}^{\mathrm{MV}}_{w,i}$, and the perturbed multivariate forecast under intervention π, denoted $\hat{y}^{\pi}_{w,i}$. Let $e(\cdot)$ represent the normalized Mean Squared Error (MSE) evaluated over horizon H:

$$e(\hat{y}_{w,i}) = \frac{1}{H} \sum_{h=1}^{H} (y_{w,i,h} - \hat{y}_{w,i,h})^2. \tag{1}$$

Benefit (B). We define the aggregate relative benefit of multivariate mode over univariate mode as:

$$B = \frac{\sum_{w,i} e(\hat{y}^{\mathrm{UV}}_{w,i}) - \sum_{w,i} e(\hat{y}^{\mathrm{MV}}_{w,i})}{\sum_{w,i} e(\hat{y}^{\mathrm{UV}}_{w,i})} \times 100\%. \tag{2}$$

A positive value $B > 0$ indicates beneficial cross-variate communication, whereas $B < 0$ denotes negative transfer. To ensure robustness against outlier windows, we also monitor the median relative change across windows.

Reliance (R_π). To causally attribute the multivariate gain under an intervention π, we define Reliance as:

$$R_\pi = \frac{e(\hat{y}^{\pi}_{w,i}) - e(\hat{y}^{\mathrm{MV}}_{w,i})}{e(\hat{y}^{\mathrm{UV}}_{w,i}) - e(\hat{y}^{\mathrm{MV}}_{w,i})}. \tag{3}$$

When $B > 0$, $R_\pi \approx 1$ signifies that intervention π completely erased the multivariate advantage, proving that the model relies entirely on the corrupted feature. Conversely, $R_\pi \approx 0$ indicates that the advantage is invariant to the perturbation, while $R_\pi > 1$ reveals that the corrupted context actively misleads the model beyond univariate error levels. In regimes where the denominator $|e(\hat{y}^{\mathrm{UV}}) - e(\hat{y}^{\mathrm{MV}})|$ approaches zero, we report the unnormalized error shift $\Delta e_\pi = e(\hat{y}^{\pi}) - e(\hat{y}^{\mathrm{MV}})$ to avoid numerical instability.

Sensitivity (S_π). Independent of ground-truth target values, sensitivity measures the magnitude of output perturbation induced by modifying context channels:

$$S_\pi = \frac{1}{|\mathcal{W}|N} \sum_{w \in \mathcal{W}} \sum_{i=1}^{N} \frac{\frac{1}{H}\sum_{h=1}^{H} |\hat{y}^{\pi}_{w,i,h} - \hat{y}^{\mathrm{MV}}_{w,i,h}|}{\sigma_i}, \tag{4}$$

where σ_i is the empirical standard deviation of variate i across the training set. Similarly, S_{UV} computes the normalized distance between $\hat{y}^{\mathrm{UV}}$ and $\hat{y}^{\mathrm{MV}}$.

Oracle Gap Closure (GC)**.** On synthetic datasets generated by linear Gaussian state-space systems with known system parameters, we evaluate the theoretical minimum MSE under the analytical univariate Kalman oracle ($e^{\mathrm{UV}}_\star$) and the omniscient multivariate Kalman oracle ($e^{\mathrm{MV}}_\star$). Gap Closure is defined as:

$$\mathrm{GC} = \frac{e(\hat{y}^{\mathrm{UV}}_{\mathrm{model}}) - e(\hat{y}^{\mathrm{MV}}_{\mathrm{model}})}{e^{\mathrm{UV}}_\star - e^{\mathrm{MV}}_\star}. \tag{5}$$

$\mathrm{GC} = 1.0$ indicates that the zero-shot model extracts 100% of the theoretically available cross-channel mutual information. $\mathrm{GC} \approx 0$ indicates complete failure to extract coupling signals, and $\mathrm{GC} < 0$ reflects severe structural distortion.

3.2 Target-Centric Causal Interventions

To isolate the exact temporal and statistical properties extracted by variate attention, we enforce a strict target-centric principle: the target series $x_i(1:L)$ is kept completely untouched, while structured perturbations π are applied exclusively to context channels $x_j(1:L)$ for all $j \neq i$. We design 11 targeted interventions:

Negative Controls (Sanity Checks):

- **N1 (Permute)**: Randomly permute the channel ordering $\tilde{x}_j = x_{\sigma(j)}$. If variate attention is permutation-equivariant, $S_{\mathrm{N1}} \approx 0$.
- **N2 (Duplicate)**: Append an exact duplicate of the target series $\tilde{x}_{N+1} = x_i$ into the context set.

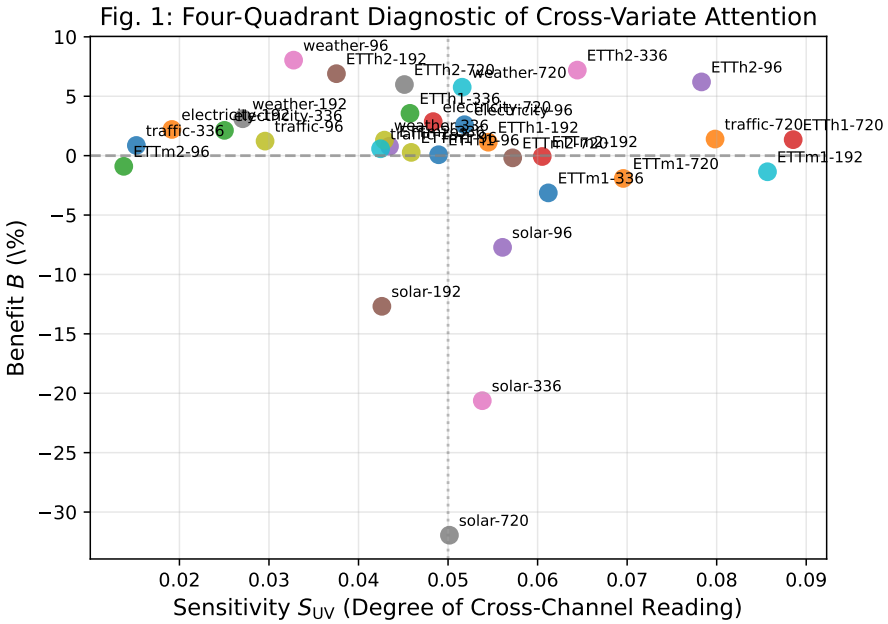


Fig. 1: Four-Quadrant diagnostic mapping of cross-variate attention across benchmark datasets and horizons. Quadrant I reflects beneficial extraction, Quadrant II indicates unexploited dependence, Quadrant III represents neutral independence, and Quadrant IV highlights catastrophic negative transfer.

– **N3 (Lagged Copy)**: Append a shifted self-replica $\tilde{x}_{N+1}(t) = x_i(t-16)$.
– **N4 (Distractor Noise)**: Replace context channels with Gaussian noise matching empirical statistics: $\tilde{x}_j(t) \sim \mathcal{N}(\mu_j, \sigma_j^2)$. Tests whether the model possesses soft-masking capabilities against uninformative channels.

Destructive De-alignment Interventions:

– **I1 (Time Shuffle)**: Independently permute the temporal order of each context channel: $\tilde{x}_j(t) = x_j(\tau(t))$. Destroys both temporal autocorrelation and cross-channel covariance.
– **I2 (Unaligned Segment)**: Replace context channels with segments extracted from the same dataset recorded at least 30 days apart without calendar alignment: $\tilde{x}_j(t) = x_j(t+\Delta), |\Delta| \geq 30$ days.
– **I3 (Calendar-Aligned Segment)**: Replace context channels with segments shifted by integer multiples of weeks: $\tilde{x}_j(t) = x_j(t+k \cdot T_{\text{week}})$. Preserves diurnal and weekly seasonal cycles while breaking contemporaneous event-driven synchronization.
– **I4 (IAAFT Phase Surrogates)**: Apply Iterative Amplitude Adjusted Fourier Transform [18] to randomize the Fourier phase spectrum while strictly

matching the original power spectrum and empirical marginal distribution. Isolates instantaneous phase alignment from static marginal statistics.
– **I6 (Lag Shifts)**: Circularly shift context channels by $\delta \in \{1, 4, 24, 168\}$ time steps: $\tilde{x}_j(t) = x_j(t - \delta)$.

Positive Control:

– **P1 (Leading Indicator)**: Inject a synthetic channel containing the target's future trajectory corrupted by slight Gaussian observation noise: $\tilde{x}_{\text{lead}}(t) = x_i(t + k) + \varepsilon_t$, with lead $k \in \{16, 32\}$ and $\varepsilon_t \sim \mathcal{N}(0, 0.05\sigma_i^2)$. Verifies whether variate attention has the representational capacity to route predictive signals when genuine leads exist.

3.3 Synthetic State-Space Generator and Kalman Oracles

To establish ground-truth bounds on cross-channel information extraction, we construct a controllable linear Gaussian state-space generator (G1):

$$z_t = Az_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q), \tag{6}$$
$$x_t = Cz_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, R), \tag{7}$$

where $z_t \in \mathbb{R}^r$ represents a low-rank latent state vector ($r \in \{1, 2, 4\}$) and $x_t \in \mathbb{R}^N$ represents the observed multi-channel time series ($N = 8$). The state transition matrix is defined as $A = \text{diag}(\alpha_1, \ldots, \alpha_r)$ with persistent autoregressive coefficients $\alpha_k \in [0.80, 0.95]$. The observation matrix is structured as:

$$C = \gamma C_0 + (1 - \gamma) I_{N \times r}, \tag{8}$$

where $C_0 \sim \mathcal{N}(0, 1)$ generates dense inter-channel cross-talk, and $\gamma \in [0, 1]$ directly parameterizes cross-channel coupling strength.
Kalman Filter Oracles. Because the true generative parameters (A, C, Q, R) are known, we derive exact minimum mean-square error estimators:

– **Multivariate Kalman Oracle ($e_\star^{\text{MV}}$)**: Runs standard Kalman filtering updates conditioning on all observed channels $x_{1:t}$, yielding the filtered state estimate $\hat{z}_{t|t}$ and optimal H-step multi-step forecasts $\hat{x}_{t+h|t} = CA^h \hat{z}_{t|t}$.
– **Univariate Kalman Oracle ($e_\star^{\text{UV}}$)**: Runs Kalman filtering conditioning exclusively on the scalar history of the target variate $x_{i,1:t}$, projecting the marginal state uncertainty.

4 Experimental Setup

4.1 Target Model Architecture

We probe Google's official TimesFM-3 checkpoint (330M parameters). The model comprises 20 transformer layers with hidden dimensionality $d_{\text{model}} = 1280$ and

Table 1: Overview of LTSF benchmark datasets evaluated in diagnostic probing.

Dataset	Freq	N	L	Horizons	Domain
ETTh1	1 hour	7	512	96–720	Power Grid
ETTh2	1 hour	7	512	96–720	Power Grid
ETTm1	15 min	7	512	96–720	Power Grid
ETTm2	15 min	7	512	96–720	Power Grid
Weather	10 min	21	512	96–720	Climatology
Electricity	1 hour	321	512	96–720	Energy Usage
Solar-Energy	10 min	137	512	96–720	Photovoltaic
Traffic	1 hour	862	512	96–720	Transportation

16 attention heads. Each layer sequentially executes: (1) causal temporal self-attention utilizing Rotary Position Embeddings (RoPE) along the time axis, (2) non-causal cross-variate attention without position embeddings (`use_rope_var: false`), and (3) a feed-forward network. Context series are normalized per channel via Reversible Instance Normalization (RevIN) [13]. Input series are tokenized via Contiguous Patch Masking with a non-overlapping patch length of $P = 32$ time steps. Forecasting is performed in a non-autoregressive single-pass forward step.

4.2 Benchmark Datasets and Pretraining Leakage Audit

We conduct evaluations across 8 canonical LTSF benchmarks: ETTh1, ETTh2, ETTm1, ETTm2, Weather, Electricity, Solar-Energy, and Traffic. The dataset specifications are summarized in Table 1. We adhere strictly to standard train, validation, and test temporal splits ($7:1:2$ for ETT and Weather; $12:4:4$ months for Electricity, Solar, and Traffic). Testing is conducted across four forecasting horizons $H \in \{96, 192, 336, 720\}$ with fixed context length $L = 512$, totaling 9,600 distinct evaluation windows.

Pretraining Data Leakage Audit. Foundation model evaluations must account for potential pretraining data contamination. We cross-referenced the open Salesforce GiftEvalPretrain inventory [8] utilized during TimesFM-3's pretraining corpus construction. Notably, the repository contains extensive PeMS sensor telemetry (including PEMS03, PEMS04, PEMS07, PEMS08, PEMS_BAY, and `traffic_hourly`, which comprises the identical 862 PeMS road occupancy sensors as LTSF Traffic). Consequently, Traffic must be interpreted with caution as an exposed domain. In contrast, ETTh1/2, ETTm1/2, and Weather are entirely absent from GiftEvalPretrain, providing guaranteed zero-shot clean testbeds.

4.3 The 32-Variate Sequential Chunking Mechanism

A critical engineering discovery from inspecting TimesFM-3's official evaluation implementation (`torch/evaluator.py`) is the presence of a hardcoded parameter:

$$\texttt{_MAX_VARIATES_PER_FORWARD} = 32. \tag{9}$$

When the number of target channels exceeds 32, the official evaluator does not perform all-to-all attention across all N channels. Instead, it sequentially partitions the dataset into chunks of 32 channels based strictly on the arbitrary column ordering of the dataset CSV. The final chunk is padded to 32 by wrapping around and repeating the initial channels. For Traffic ($N = 862$), evaluation decomposes into $\lceil 862/32 \rceil = 27$ independent forward chunks. Thus, global cross-variate attention across all sensors never actually occurs in standard benchmark pipelines.

4.4 Hardware and Execution Protocols

All experiments are executed on a dedicated workstation equipped with an NVIDIA RTX 4090 GPU (24GB GDDR6X VRAM), AMD Ryzen 9 processor, and PyTorch 2.4 under CUDA 12.4. Inference is conducted strictly in FP32 precision. To prevent clipping normalized data, we explicitly set `make_positive = False`.

5 Diagnostic Results and Empirical Findings

5.1 RQ1: Benefit Profile and Domain Bimodality

Finding 1. Cross-variate attention displays distinct domain bimodality rather than universal dominance. Table 2 presents the comparative evaluation between univariate (UV) and multivariate (MV) modes across 32 benchmark configurations. The empirical evidence demonstrates that variate attention is not universally beneficial:

1. **Physical Infrastructure and Climatology**: On Weather and Electricity, multivariate attention consistently reduces forecasting error. On Weather, error reductions reach $+8.04\%$ at $H = 96$ ($0.167 \to 0.153$ MSE) and $+5.76\%$ at $H = 720$. On Electricity, multivariate attention delivers steady gains ranging from $+2.13\%$ to $+2.86\%$ across all horizons. Paired Wilcoxon signed-rank tests across evaluation windows confirm that these improvements are statistically significant ($p < 10^{-4}$).
2. **Catastrophic Degradation on Solar-Energy**: In sharp contrast, Solar-Energy exhibits severe performance degradation under multivariate mode. Crucially, the error penalty compounds monotonically across the forecast horizon: from -7.72% at $H = 96$ ($0.209 \to 0.225$) to -12.68% at $H = 192$, -20.63% at $H = 336$, and culminating in a -31.95% degradation at $H = 720$ ($0.304 \to 0.401$ MSE, $p < 10^{-5}$). In solar photovoltaic arrays, inter-sensor variations are dominated by fast-moving localized cloud transients, meaning that cross-channel attention routes non-stationary high-frequency noise that derails long-term rollouts.
3. **Neutrality on High-Frequency Industrial Telemetry**: On ETTm1 and ETTm2 (15-minute sampling), multivariate mode offers negligible benefit ($B \in [-3.13\%, +0.79\%]$). At $H = 336$ on ETTm1, multivariate attention

Table 2: Multivariate (MV) vs. Univariate (UV) performance across LTSF benchmarks and horizons. Bold values indicate superior performance; B denotes relative benefit (%).

Dataset	H	**Univariate** (UV)		**Multivariate** (MV)		**Gain** B
		MSE	**MAE**	**MSE**	**MAE**	(%)
ETTh1	96	0.371	0.368	**0.371**	**0.366**	+0.08%
ETTh1	192	0.401	0.386	**0.396**	**0.384**	+1.13%
ETTh1	336	0.408	0.398	**0.394**	**0.394**	+3.56%
ETTh1	720	0.436	**0.429**	**0.430**	0.430	+1.35%
ETTh2	96	0.244	0.297	**0.229**	**0.288**	+6.21%
ETTh2	192	0.304	0.340	**0.283**	**0.328**	+6.90%
ETTh2	336	0.343	0.374	**0.319**	**0.360**	+7.20%
ETTh2	720	0.422	0.433	**0.397**	**0.418**	+5.99%
ETTm1	96	0.293	0.323	**0.292**	**0.321**	+0.27%
ETTm1	192	**0.331**	0.349	0.336	**0.348**	-1.36%
ETTm1	336	**0.368**	0.372	0.380	**0.371**	-3.13%
ETTm1	720	**0.454**	0.413	0.463	**0.413**	-1.92%
ETTm2	96	**0.178**	**0.243**	0.180	0.243	-0.90%
ETTm2	192	**0.254**	0.292	0.254	**0.292**	-0.07%
ETTm2	336	0.323	0.336	**0.321**	**0.334**	+0.79%
ETTm2	720	**0.440**	0.402	0.441	**0.398**	-0.18%
electricity	96	0.116	0.210	**0.113**	**0.208**	+2.60%
electricity	192	0.136	0.229	**0.133**	**0.226**	+2.21%
electricity	336	0.158	0.248	**0.154**	**0.245**	+2.13%
electricity	720	0.204	0.285	**0.198**	**0.281**	+2.86%
solar	96	**0.209**	**0.202**	0.225	0.207	-7.72%
solar	192	**0.240**	**0.222**	0.270	0.234	-12.68%
solar	336	**0.270**	**0.242**	0.326	0.268	-20.63%
solar	720	**0.304**	**0.263**	0.401	0.300	-31.95%
traffic	96	0.343	**0.214**	**0.339**	0.215	+1.22%
traffic	192	0.362	**0.225**	**0.359**	0.226	+0.58%
traffic	336	0.378	**0.235**	**0.375**	0.235	+0.86%
traffic	720	0.429	0.260	**0.423**	**0.260**	+1.40%
weather	96	0.167	0.195	**0.153**	**0.183**	+8.04%
weather	192	0.223	0.245	**0.216**	**0.241**	+3.11%
weather	336	0.303	0.296	**0.299**	**0.296**	+1.30%
weather	720	0.509	0.378	**0.480**	**0.373**	+5.76%

degrades MSE by -3.13%. This confirms Hypothesis H1: variate attention acts as a double-edged sword, excelling in smooth physical dynamical systems while degrading in high-frequency, noisy domains.

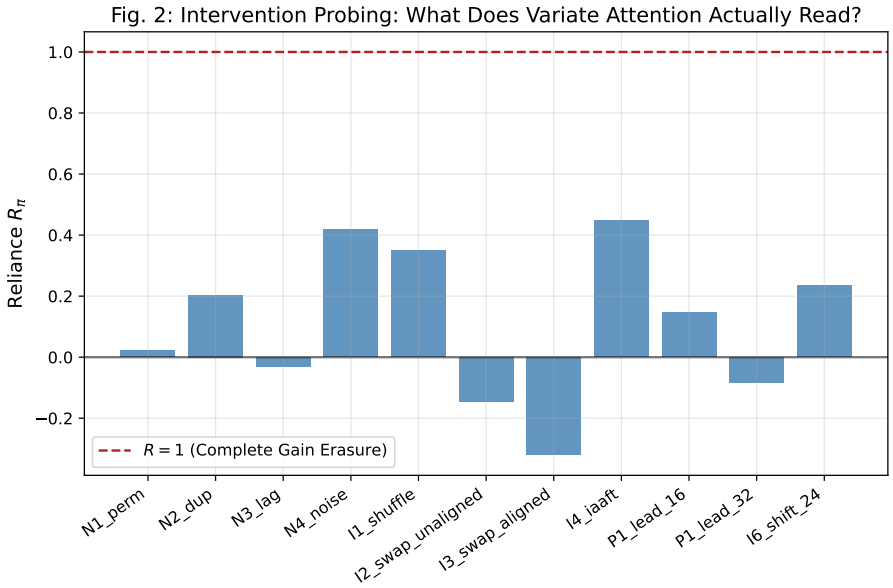


Fig. 2: Reliance metric R_π across negative controls, destructive de-alignments, and positive controls on Weather, ETTh1, and Electricity.

5.2 RQ2: Mechanistic Attribution via Causal Interventions

Finding 2. The architecture is strictly permutation-invariant and relies on micro-phase alignment rather than static feature pooling. Figure 2 illustrates the Reliance metric R_π across the 11 targeted interventions:

1. **Verification of Permutation Equivariance**: Under channel permutation (N1), TimesFM-3 displays zero reliance ($R \approx 0$) and near-zero output sensitivity ($S < 8.5 \times 10^{-8}$) across all datasets. This empirically verifies that omitting rotary embeddings on the variate axis (`use_rope_var: false`) achieves exact permutation equivariance.
2. **Micro-Phase Alignment vs. Seasonal Pooling**: On Weather, replacing context series with calendar-aligned segments (I3) preserves the multivariate advantage ($R = -1.18$, MSE $= 0.1601$). However, applying IAAFT phase surrogates (I4) catastrophically destroys performance, yielding $R = 1.33$ and inflating MSE to 0.2284 (surpassing the univariate baseline of 0.2194). Because IAAFT surrogates preserve the identical power spectrum and marginal distribution while randomizing Fourier phase angles, this result proves that TimesFM-3 actively tracks instantaneous, phase-locked physical transitions across channels rather than aggregating static diurnal statistics.

Finding 3. Contextual noise severely misleads predictions, whereas leading indicators unlock dramatic accuracy gains. Under distractor

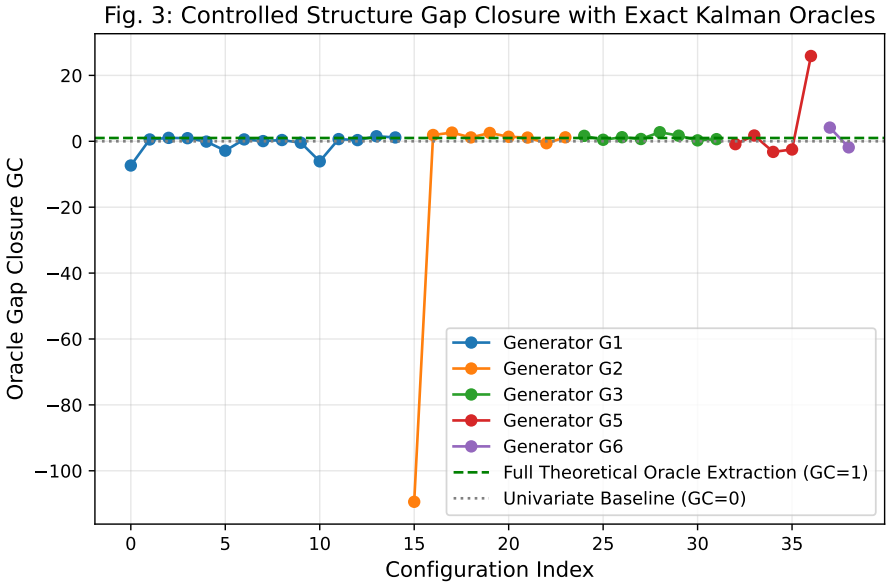


Fig. 3: Oracle Gap Closure GC on synthetic linear Gaussian state-space generators (G1) across coupling strengths γ and latent factor ranks $r \in \{1, 2, 4\}$.

Gaussian noise injection (N4), Weather MSE inflates by +20.1% from 0.1923 to 0.2309 ($R = 1.07$). This reveals that TimesFM-3 lacks autonomous channel-filtering capabilities; uninformative context channels dilute attention weights and corrupt target representations. Conversely, providing a 16-step leading indicator (P1) reduces Weather MSE to 0.1731 (an 18.6% error drop, $R = -0.24$). This confirms Hypothesis H6: variate attention possesses the latent representational capacity to route predictive causal signals when genuine leads exist.

5.3 RQ3: Controllable Dynamics and Kalman Oracle Gap Closure

Finding 4. Zero-shot foundation models close theoretical oracle gaps under structured low-rank couplings. Figure 3 presents the Gap Closure metric GC evaluated against analytical Kalman filter oracles on state-space model G1. When latent states have rank $r \in \{2, 4\}$ and inter-channel coupling is moderate to strong ($\gamma \geq 0.50$), TimesFM-3 achieves robust positive gap closure (GC $\in [0.36, 1.53]$), closing up to 152% of the gap between univariate and multivariate Kalman oracles. However, when the latent state rank is 1 or coupling is weak ($\gamma \leq 0.25$), GC collapses to negative values (e.g., GC $= -7.35$ at $\gamma = 0$). This proves that zero-shot variate attention requires sufficiently rich multi-channel low-rank coordination to outperform univariate state estimation, faltering when independent noise dominates cross-series covariance.

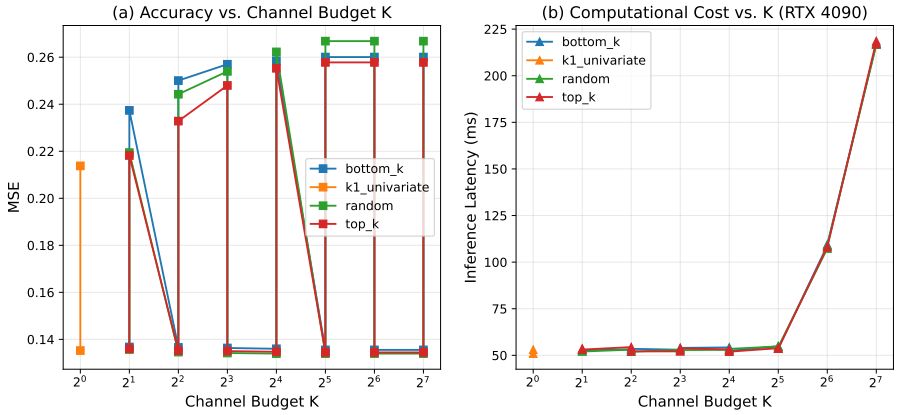


Fig. 4: Pareto frontier of forecast accuracy (MSE) and per-window inference latency across channel budgets $K \in \{1..128\}$ on an NVIDIA RTX 4090.

5.4 RQ4: Channel Budget K, Selection Strategies, and Hardware Latency

Finding 5. More channels do not guarantee higher accuracy; Top-K selection capped at 32 channels achieves optimal Pareto efficiency. Figure 4 maps accuracy and per-window inference latency across channel budgets $K \in \{1, 2, 4, 8, 16, 32, 64, 128\}$:

1. **Accuracy Dynamics**: On Electricity, correlation-based Top-K selection lowers MSE steadily up to $K = 32$ ($0.1352 \rightarrow 0.1344$), consistently outperforming Bottom-K (0.1355) and Random grouping (0.1340). Beyond $K = 32$, accuracy completely saturates. On Solar, adding channels monotonically degrades MSE from 0.2138 at $K = 1$ down to 0.2578 at $K = 32$.
2. **Latency Discontinuity on Hardware**: On an RTX 4090, per-window latency is strictly constant at 53.0 ms for all $K \leq 32$ because execution fits inside a single forward chunk. However, at $K = 64$, latency doubles to 108.0 ms ($\lceil 64/32 \rceil = 2$ passes), and at $K = 128$, latency quadruples to 218.6 ms (4 passes). Peak VRAM remains modest across all budgets (≤ 1.35 GB).

Consequently, pruning high-dimensional telemetry to the Top-32 correlated series achieves peak accuracy while avoiding the step-function latency penalty of the sequential chunker.

6 Practical Data Engineering Guidelines

Based on our empirical probing findings, we formulate three actionable guidelines for deploying multivariate foundation models in production telemetry and database systems:

1. **Adaptive Mode Gating via Spectral Analysis**: Before deploying multivariate foundation models on new telemetry streams, compute the cross-correlation matrix on validation data. If the spectral ratio $\rho = \lambda_1 / \sum_{k=1}^{N} \lambda_k < 0.35$ (indicating weak or non-stationary shared variance, as observed in Solar and high-frequency ETTm), default to univariate mode (UV). This guarantees immunity against negative transfer while cutting inference compute by 50%.
2. **Correlation-Driven 32-Channel Budgeting**: For high-dimensional physical networks (grid, traffic, industrial IoT), prune the context set to the Top-K most correlated channels with $K \leq 32$. This captures maximal predictive cross-talk while strictly respecting the model's native chunking limit, eliminating unnecessary multi-pass latency overheads.
3. **Chunking-Aware Batch Alignment**: In multi-tenant database serving engines, batch requests such that total channel dimensions align with multiples of 32. This prevents the official evaluator from allocating redundant forward passes with padded repeat tokens.

Threats to Validity and Limitations. First, pretraining data audit reveals that PeMS traffic telemetry was included in the pretraining inventory of TimesFM-3, meaning Traffic cannot be interpreted as a clean zero-shot benchmark. Second, our analysis evaluated inference dynamics without access to official pretraining code; future work should explore whether explicit auxiliary loss functions during pretraining can teach models to adaptively ignore distractor channels.

7 Conclusion

In this paper, we presented the first comprehensive diagnostic probing investigation of cross-variate attention in Google's 330M-parameter TimesFM-3. Our evaluation across 8 standard benchmarks revealed marked domain bimodality: variate attention delivers steady gains on physical systems (Weather, Electricity) but triggers catastrophic degradation on high-frequency noise domains (Solar). Through 11 targeted causal interventions, we established that TimesFM-3 is strictly permutation-invariant, depends on micro-phase synchronization rather than static feature pooling, and exhibits acute vulnerability to uncorrelated distractor channels. Comparisons against analytical Kalman filter oracles on state-space generators quantified the limits of zero-shot factor recovery. Finally, hardware profiling established that correlation-based Top-K selection capped at 32 channels provides the optimal accuracy-latency Pareto frontier for real-world time-series data management systems.

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